

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE AI 2024	Date:	18-08-2026

Code of Conduct

I _Y SRAVAN KUMAR (192472289) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____SRAVAN_____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation

		structure, visuals, and terminology			and difficult to understand
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Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	3	3	3	3	4	16	76
2	3	4	3	3	3	16	
3	4	3	3	3	3	16	
4	3	3	4	3	3	16	

Faculty In-charge	Course Coordinator	Program Director
Signature: Dr.T.Kumaragurubaran_	Signature: _____	Signature: _____
Date: ____18-08-2026__	Date: _____	Date: _____

1.Frame Semantics in Banking Customer-Service Systems

1. Semantic Frame Analysis

Banking Request	Action	Participants / Roles
Transfer ₹20,000 from savings to son's account	TRANSFER	Source: Savings Account, Amount: ₹20,000, Destination: Son's Account, Agent: Customer
Block my debit card	BLOCK	Object: Debit Card, Agent: Customer
Send transaction statement to email	SEND	Object: Transaction Statement, Destination: Email, Agent: Customer
Increase daily withdrawal limit	MODIFY	Object: Withdrawal Limit, Agent: Customer

Relationship: The action represents what the customer wants to do, while the semantic roles identify the entities involved in that action.

2. Incorrect Semantic Interpretation

Query B4 is incorrect.

- Actual Intent: Increase withdrawal limit.
- System Interpretation: Decrease withdrawal limit.

The action "increase" means to raise the limit, while "decrease" means to lower it. Therefore, the system has misunderstood the intended action.

3. Effect of Incorrect Semantic Roles

Incorrect semantic-role identification can cause serious banking errors.

- Wrong source/destination → money could be transferred to the wrong account.
- Wrong amount → incorrect transaction value.
- Wrong action → a limit could be decreased instead of increased.
- Wrong object → the wrong card or account could be modified.

Therefore, semantic errors can result in financial loss, security problems, and customer dissatisfaction.

4. Recommended Frame-Based Approach

A reliable banking chatbot should:

1. Identify the action – Transfer, Block, Send, Modify, etc.
2. Identify semantic roles – Customer, account, amount, destination, object, etc.
3. Validate all important entities before performing an operation.
4. Check the user's intent against banking rules.
5. Ask for confirmation before high-risk transactions.
6. Use context from previous messages to resolve ambiguous requests.

2. First-Order Predicate Calculus for Smart Agriculture

1. Field Observations in First-Order Predicate Calculus

F1:

- Dry(F1)
- IrrigationAvailable(F1)
- Rice(F1)

F2:

- Wet(F2)
- IrrigationAvailable(F2)
- Wheat(F2)

F3:

- Dry(F3)
- IrrigationUnavailable(F3)
- Maize(F3)

F4:

- Wet(F4)
- IrrigationUnavailable(F4)

- Rice(F4)

2. Fields Requiring Irrigation

Rule:

For F1:

Therefore:

Then:

So:

F2: Wet $\rightarrow$ Does not need water.

F3: Dry, but irrigation is unavailable $\rightarrow$ cannot be irrigated.

F4: Wet $\rightarrow$ Does not need water.

Final Result

	Field Needs Water	Irrigate
F1	Yes	Yes
F2	No	No
F3	Potentially Yes	No – unavailable
F4	No	No

3. F1 vs F3

Yes, F1 and F3 should be treated differently.

- F1: Dry + irrigation available $\rightarrow$ Irrigate
- F3: Dry + irrigation unavailable $\rightarrow$ Do not irrigate

Although both fields are dry, the irrigation availability changes the controller's action.

4. Is Predicate Logic Alone Sufficient?

No. Predicate logic works well with clear, definite sensor information, but real agricultural environments contain uncertainty.

For example:

- Sensors may provide inaccurate readings.
- Soil moisture may be partially measured.
- Weather conditions can change.
- Irrigation availability may change.
- Sensor data may be missing.

Therefore, predicate logic should be combined with fuzzy logic, probability, machine learning, or sensor-confidence models to make better decisions under uncertain or incomplete information.

3. Word Sense Disambiguation in a Travel Recommendation System

1. Intended Sense of Ambiguous Terms

Query	Intended Sense	Reason
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages.
Safari booking	Wildlife tour	User selected wildlife resort packages.
Java vacation	Java island	User viewed hotels in Bali and Java.
Cruise package	Ship journey	User viewed Mediterranean ship packages.
Safari download for Windows Web browser		User clicked a browser download page.

2. Lexical and Contextual Cues

- Amazon: Words like *tour*, *rainforest*, *trekking* indicate the geographical sense.
- Safari: *Booking*, *wildlife*, *resort* indicate an animal-tour sense; *download*, *Windows*, *browser* indicate the software sense.
- Java: *Vacation*, *hotels*, *Bali* indicate the Indonesian island.
- Cruise: *Package*, *Mediterranean*, *ship* indicate a sea journey.

Lexical cues are the words surrounding the ambiguous word, while contextual cues include the user's search history and interactions.

3. Different Interpretations for Different Users

The same word can have different meanings depending on the user's query context, interests, location, previous searches, and interactions.

For example:

- User A: "Safari wildlife booking" → Safari wildlife tour.
- User B: "Safari download Windows" → Safari web browser.

Therefore, WSD should consider individual user context instead of relying only on the ambiguous word.

4. Industrial-Scale WSD Strategy

A travel recommendation system can use:

1. Query Context – Analyze surrounding words and phrases.
2. User History – Use previous searches, viewed hotels, and bookings.
3. Embeddings – Use models such as BERT to understand semantic similarity.
4. Click Behaviour – Learn from pages and packages clicked by users.

5. Candidate Sense Ranking – Assign scores to each possible meaning.
6. Feedback Loop – Continuously update predictions using new user interactions.

Example

"Safari download for Windows"

Safari + download + Windows → Browser sense

"Safari booking wildlife resort"

Safari + booking + wildlife + resort → Wildlife-tour sense

This approach provides more accurate and personalized travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. Effect of Syntactic Structure on Semantic Roles

Syntactic structure helps determine who performs an action, what is affected, and who receives it.

- Active voice: The grammatical subject is usually the Agent.
 - *The supplier delivered the equipment.*
 - Supplier → Agent
 - Equipment → Object
 - Buyer → Recipient
- Passive voice: The grammatical subject is usually the Object/Patient, while the Agent is often introduced by "by".
 - *The contract was terminated by the company.*
 - Contract → Object/Patient
 - Company → Agent

2. Incorrect Semantic-Role Assignments

There are two incorrect interpretations.

Sentence 2

"The buyer received the equipment from the supplier."

Given interpretation:

- Buyer = Agent
- Equipment = Object
- Supplier = Recipient

Correct:

- Buyer = Recipient
- Equipment = Object

- Supplier = Source/Agent

Sentence 4

"The contract was terminated by the company."

Given interpretation:

- Contract = Agent
- Company = Object

Correct:

- Company = Agent
- Contract = Object/Patient

Sentence 1 and Sentence 3 are correctly interpreted.

3. Effect of Confusing Subject with Agent

A grammatical subject is not always the semantic Agent, especially in passive sentences.

For example:

The contract was terminated by the company.

The grammatical subject is contract, but the actual Agent is company.

If a legal NLP system treats every subject as an Agent, it may incorrectly extract:

Agent = Contract

instead of:

Agent = Company

This can cause serious errors in:

- Contract analysis
- Liability identification
- Legal information extraction
- Compliance checking
- Dispute analysis

4. Recommended Syntax-Driven Architecture

A reliable legal NLP system can use the following pipeline:

Legal Document → Tokenization → POS Tagging → Dependency Parsing → Voice Detection → Semantic Role Labeling → Entity Linking → Relation Extraction

Key Components

1. POS Tagging – Identifies grammatical categories.
2. Dependency Parsing – Finds subject, object, and modifiers.
3. Voice Detection – Distinguishes active and passive constructions.
4. Semantic Role Labeling – Identifies Agent, Patient, Recipient, Source, etc.

5. Prepositional Phrase Analysis – Interprets phrases such as *"by the company"* and *"from the supplier"*.
6. Coreference Resolution – Connects references such as *"it"*, *"they"*, and *"the party"* to the correct entities.
7. Long-Distance Dependency Handling – Connects semantic roles even when important information is separated by clauses.

Final Architecture

Syntax + Dependency Parsing + Semantic Role Labeling + Coreference Resolution provides a robust approach for accurately extracting legal relationships from both active and passive sentences.